

Day 04: Inference for Prediction

Name: _____

Last time, and today

Last time: the sampling distribution for β_i 's, confidence intervals and hypothesis tests for β_i 's, both by hand and in R.

Today: inference for *predictions*.

Two types of predictions

1. Predicting the **mean response** at a specific value of x – e.g. the *average* number of riders on a 62 degree day.
2. Predicting the response for a **specific future observation** – e.g. the number of riders *tomorrow* (forecast to be 62 degrees).



Tip

Think of two additional real-world examples of each type of prediction.

Inference for prediction

Best estimate: $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$

General interval formula:

$$\text{estimate} \pm t^* \times SE$$

- $\hat{y}_0$ is our estimate
- use a t -distribution with $df = n - 2$ to find t^*

We'll need a *different* SE depending on whether we're building:

- a confidence interval for the mean response $\hat{\mu}(Y \mid X_0)$, or
- a prediction interval for an individual response $Y \mid X_0$

Standard errors

$$SE(\hat{\mu}(Y | X)) = \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

$$SE(\hat{y}) = \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

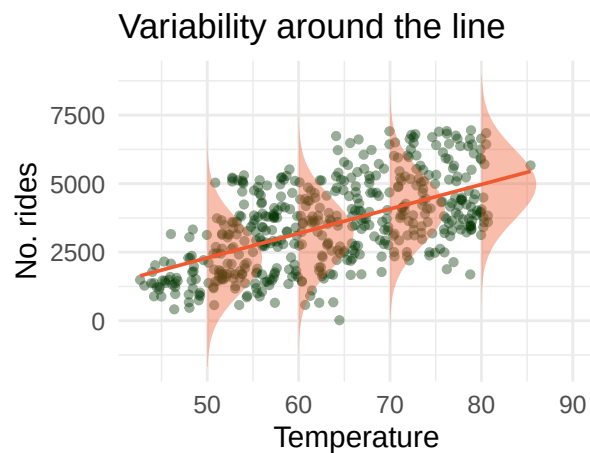
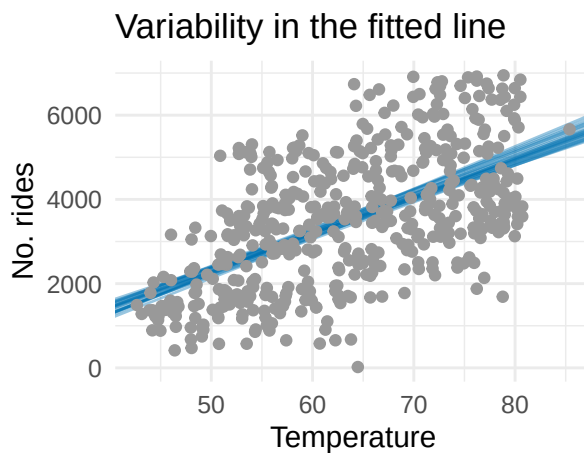
Comparing the two formulas: the prediction interval SE has an extra “+1” under the square root, so it will always be _____ (wider/narrower) than the confidence interval SE.

As x_0 moves farther from $\bar{x}$, both standard errors _____ (grow/shrink), because $(x_0 - \bar{x})^2$ grows.

Where the extra variability comes from

$$SE(\hat{y}) = \sqrt{\underbrace{\hat{\sigma}^2}_{\text{variability in } \varepsilon_i\text{'s}} + \underbrace{\frac{\hat{\sigma}^2}{n} + \frac{\hat{\sigma}^2(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}_{\text{variability in } \hat{\beta}_i\text{'s}}}$$

The first term is the leftover noise around the line at any x ; the rest comes from not knowing the true line exactly.



R: point estimate

`predict()` quickly calculates $\hat{y}$ for a given x (or a vector of x 's):

```
predict(bikes_lm, newdata = data.frame(temp_actual = 52))
```

```
      1  
1 2471.388
```

```
predict(bikes_lm, newdata = data.frame(temp_actual = c(52, 62, 72, 82)))
```

```
      1      2      3      4  
1 2471.388 3363.727 4256.066 5148.405
```

R: intervals

The `interval` argument specifies which type of interval you want:

```
predict(bikes_lm, newdata = data.frame(temp_actual = 52), interval = "confidence")
```

```
      fit      lwr      upr  
1 2471.388 2301.879 2640.898
```

```
predict(bikes_lm, newdata = data.frame(temp_actual = 52), interval = "prediction")
```

```
      fit      lwr      upr  
1 2471.388 -65.1291 5007.906
```

Notice the prediction interval is _____ than the confidence interval, exactly as the SE formulas predicted.

R: confidence level and SEs

The `level` argument sets the confidence level (default 0.95):

```
predict(bikes_lm, newdata = data.frame(temp_actual = 52), interval = "confidence", level = .95)
```

```
      fit      lwr      upr  
1 2471.388 2301.879 2640.898
```

```
predict(bikes_lm, newdata = data.frame(temp_actual = 52), interval = "prediction", level = .99)
```

```
      fit      lwr      upr
1 2471.388 -866.8508 5809.628
```

Adding `se.fit = TRUE` returns the standard error needed for “by hand” calculations:

```
predict(bikes_lm, newdata = data.frame(temp_actual = 52), se.fit = TRUE)
```

```
$fit
```

```
      1
2471.388
```

```
$se.fit
```

```
[1] 86.2756
```

```
$df
```

```
[1] 498
```

```
$residual.scale
```

```
[1] 1288.134
```

Try it

Using `$fit` and `$se.fit` above along with the appropriate t^* , reconstruct the 95% confidence interval for the mean response at `temp_actual = 52` by hand, and check it against the `predict()` output.